Pseudolabel-Based Unreliable Sample Learning for Semi-Supervised Hyperspectral Image Classification

Huaxiong Yao, Renyi Chen, Wenjing Chen, Hao Sun, *Member, IEEE*, Wei Xie, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— Recently, pseudolabel-based deep learning methods have shown excellent performance in semi-supervised hyperspectral image (HSI) classification. These methods usually select high-confidence unlabeled samples to help optimize backbone classification networks. However, a large number of remaining low-confidence unlabeled samples, which contain rich land-covers information, are underused. In this article, we propose a pseudolabel-based unreliable sample learning (PUSL) method to fully exploit low-confidence unlabeled samples for semi-supervised HSI classification. First, to avoid overfitting the spatial distribution of labeled samples, we build a position-free transformer (PFT) as the backbone classification network. Second, PFT is initially trained with labeled samples in a supervised learning manner to obtain an initial classifier, which is then used to split unlabeled samples into reliable and unreliable unlabeled samples based on the predicted confidence. Third, reliable unlabeled samples participate in training along with labeled samples. Finally, unreliable unlabeled samples are treated as negative samples for the corresponding categories to improve the discrimination of PFT in a contrastive learning paradigm. Extensive experiments on three HSI datasets demonstrate that PUSL outperforms the compared methods.**

***Index Terms*— Contrastive learning, hyperspectral image (HSI) classification, pseudolabel, semi-supervised learning.**

I. Introduction

REMOTE sensing hyperspectral image (HSI) can collect rich spectral signatures and complex spatial distributions of land covers from observed scenes [1], [2], [3], which have been widely used in marine protection, environmental monitoring, and mineral resource exploration [4], [5], [6]. HSI classification refers to predicting the categories of pixels based on spectral signatures and spatial distributions contained in HSI [7], [8], [9]. Due to the powerful feature learning ability, deep-learning-based methods have been widely used in HSI classification [10], [11]. However, most of the existing methods are supervised classification methods, which rely on a large number of labeled samples for supervised training [12], [13], [14]. For remote sensing HSI, labeling a large number of samples is a time-consuming and labor-intensive task.

Many works have been devoted to the study of training models with a small number of labeled samples for HSI classification, including transfer learning-based methods [15], [16], [17], generative adversarial learning-based methods [18], [19], and few-shot learning-based methods [20], [21]. The transfer learning-based methods focus on transferring the knowledge learned from the source domains with sufficient labeled samples to the target domain to improve classification accuracies [15], [17], [22]. The generative adversarial learning-based methods strive to generate as realistic samples as possible for training [18], [19]. The few-shot learning-based methods usually focus on studying the feature distribution differences between different categories [20], [23]. However, these works do not fully use massive unlabeled samples for training, which are easy to obtain and can provide plentiful land-cover information.

Recently, semi-supervised deep learning methods, which focus on exploring a small number of labeled samples and massive unlabeled samples to optimize network parameters, have been widely studied in HSI classification [24], [25], [26]. These methods can be categorized into reconstruction-based methods [24], [27], graph-based methods [28], [29], and pseudolabel-based methods [26], [30]. The reconstruction-based methods usually use autoencoders to learn feature representations on unlabeled samples in an unsupervised learning manner, and then use labeled samples to fine-tune network parameters in a supervised manner [24]. However, the features extracted by autoencoders mainly focus on reconstructing the original HSI and are not optimal for classification. The graph-based methods represent the topological information among labeled and unlabeled samples by constructing an adjacency matrix to capture complex spectral–spatial information [28], [29]. However, detailed spatial structures of HSI are difficult to be fully represented by graph structures [31]. The pseudolabel-based methods first train a classifier with labeled samples to obtain the predicted confidence of unlabeled samples under different categories, and then select the categories with high

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Huaxiong Yao, Renyi Chen, Hao Sun, and Wei Xie are with the Hubei Provincial Key Laboratory of Artificial Intelligence and Smart Learning, the School of Computer Science, and the National Language Resources Monitoring and Research Center for Network Media, Central China Normal University, Wuhan 430079, China (e-mail: haosun@ccnu.edu.cn; xw@mail.ccnu.edu.cn).

Wenjing Chen is with the School of Computer Science, Hubei University of Technology, Wuhan 430068, China.

Xiaoqiang Lu is with the College of Physics and Information Engineering, Fuzhou University, Fuzhou 350002, China.

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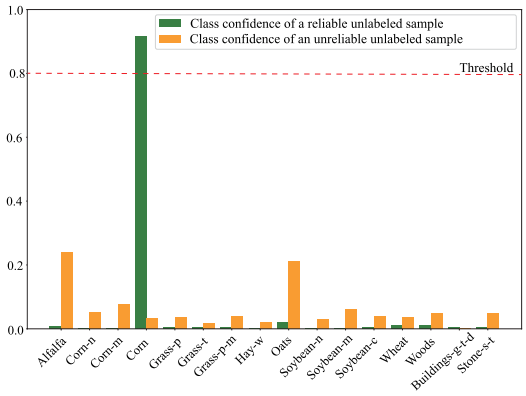

Fig. 1. Examples of the predicted confidence on different categories for a reliable unlabeled sample and an unreliable unlabeled sample.

predicted confidence as pseudolabels for unlabeled samples to optimize network parameters [26], [32].

The pseudolabel-based methods usually regard unlabeled samples whose class confidence is higher than a preset threshold as reliable unlabeled samples, and otherwise as unreliable unlabeled samples. As shown in Fig. 1, since the class confidence of the unlabeled sample in the "Corn" category is greater than the preset threshold, this sample can be regarded as a reliable unlabeled sample and the "Corn" is used as the pseudolabel of this sample. If the class confidence of an unlabeled sample in all the categories is lower than the preset threshold, this sample is regarded as an unreliable unlabeled sample. The existing pseudolabel-based methods usually exploit labeled samples and reliable unlabeled samples for training, while unreliable unlabeled samples are discarded [30], [32]. However, these unreliable unlabeled samples contain rich spectral–spatial information. Using only reliable unlabeled samples will cause unreliable unlabeled samples to never participate in model training, resulting in the underutilization of unlabeled samples.

A crude solution to using unreliable unlabeled samples is to directly use the predictions with the maximum confidence as pseudolabels for model training. However, in this case, these pseudolabels are likely to be incorrect, which will confuse the classification network and lead to poor classification performance. As shown in Fig. 1, for an unreliable unlabeled sample, although the classification network will be confused in the "Alfalfa" and "Oats" categories, this classification network is sure that this unreliable unlabeled sample does not belong to the "Buildings-g-t-d" category. Based on this phenomenon, this article proposes to use a contrastive learning paradigm to exploit unreliable unlabeled samples for training [33]. In the proposed method, this unreliable unlabeled sample can be used as a negative sample of the "Buildings-g-t-d" category to provide contrastive information to improve the model discrimination.

This article proposes a pseudolabel-based unreliable sample learning (PUSL) method to fully exploit unreliable unlabeled samples for semi-supervised HSI classification. First, a simple position-free transformer (PFT) is designed as the backbone classification network of PUSL. The existing transformer-based methods generally need to encode the spatial position information of hyperspectral pixels into spectral–spatial features for classification [34], [35]. However, when the number of labeled samples is small, their spatial distribution diversity is small, which can make these methods easy to overfit the spatial distributions of labeled samples. The proposed PFT removes the position encoding (PE) to reduce the impact of complex spatial distributions. Second, to improve the reliability of predicted class confidence for unlabeled samples, labeled samples are used to preliminarily train PFT to obtain an initial classification network. This initial classification network is further used to obtain the reliable and unreliable unlabeled samples. Third, reliable unlabeled samples participate in model training along with labeled samples. Finally, the proposed PUSL proposes to exploit the contrastive learning paradigm to regard unreliable unlabeled samples as negative samples to provide additional contrastive information for the corresponding categories to improve model discrimination. In addition, a categorywise memory bank strategy is used to allocate a fixed-size queue for each category for avoiding the imbalance of candidate negative samples in a minibatch during training. Experiments on three public HSI datasets demonstrate the superiority of PUSL.

The contributions of this article mainly include the following.

1) To fully exploit unlabeled samples, we propose a PUSL method for semi-supervised HSI classification. Unreliable unlabeled samples are exploited to improve model discrimination by contrastive learning.
2) To avoid overfitting the spatial distribution of labeled samples, we construct a PFT network as a backbone to capture discriminative spectral–spatial features by removing the PE module.
3) To avoid the imbalance of the number of candidate negative samples in different categories, a categorywise memory bank strategy is adopted to allocate a fixed-size queue for each category to store negative samples.

The rest of this article is organized as follows. Section II briefly introduces recent related methods. Section III details the proposed PUSL. Section IV describes experiments. Finally, the conclusion and future works are provided in Section V.

II. Related Work

A. Semi-Supervised HSI Classification

Semi-supervised learning can effectively alleviate the problem of dependence on the number of labeled samples [36], [37], [38]. Recent semi-supervised deep-learning-based methods can be roughly categorized into reconstruction-based methods, graph-based methods, and pseudolabel-based methods.

Typical reconstruction-based methods use autoencoders to encode high-dimensional inputs into low-dimensional semantic features [24], [27]. For example, Jia et al. [24] proposed to use an autoencoder to capture the distribution information among unlabeled samples of different categories, and then fine-tuned the network parameters with labeled samples. Sellami et al. [39] first reduced the feature dimension of HSI through redundancy criterion, and then adopted a

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3-D convolution-based encoder–decoder network to extract spectral–spatial features. However, the autoencoders mainly focus on reconstructing the original HSI, and the features extracted by autoencoders may not be optimal for classification.

The graph-based methods represent topological information of HSI through an adjacency matrix [28], [40]. For example, Ding et al. [28] combined the autoregressive moving average filter and graph neural network to capture global spatial information of HSI for semi-supervised HSI classification. Yang et al. [40] proposed a dynamic multiscale pixel-level map update region-level map structure to extract spectral–spatial information. However, detailed spatial structures of HSI are difficult to be fully represented by graph structures [31].

The pseudolabel-based methods first train an initial classifier on labeled samples, which is further used to obtain the predicted class confidence of unlabeled samples. Then, high-confidence unlabeled samples are selected together with labeled samples to optimize network parameters [32], [41], [42]. Seydgar et al. [32] proposed an uncertainty-aware model to generate reliable pseudolabels and adopted an uncertainty estimation module to improve the quality of pseudolabels. Kang et al. [41] proposed to obtain the class confidence of samples through edge-preserving features and morphological attribute profiles, and then selected high-confidence samples through the decision function for training. However, the existing pseudolabel-based methods usually use the class confidence higher than the preset threshold as pseudolabels, which leads to the phenomenon that a large number of low-confidence unlabeled samples are underused.

B. Attention Mechanism and Transformer

The attention mechanism can effectively make the model focus on extracting key features [43], [44], which has been widely used in HSI classification [45], [46], [47], [48]. Li et al. [45] proposed a dual-channel attention mechanism to improve the learning ability of HSI spectral features. Zheng et al. [46] used an attention mechanism to capture the spatial correlation between pixels. Sun et al. [48] designed a spectral–spatial attention network to extract spectral–spatial features from key regions. Liu et al. [49] proposed a novel central attention network to calculate the correlation between the center pixel and adjacent pixels. Owing to the powerful learning capability of transformer, many works have attempted to construct various transformer networks suitable for HSI classification. Deng et al. [50] proposed a hybrid model based on the transformer and convolutional neural network. He et al. [51] proposed a semi-supervised generative adversarial network combined with a vision transformer (ViT) to reduce the impact of class imbalance. Sun et al. [52] used the shallow features extracted by a convolutional neural network as the input of the transformer to further capture deep spectral–spatial information of HSI. Xie et al. [53] proposed a semantic spatial-spectral feature fusion transformer (S3FFT) network that uses the information extracted by spatial attention and channel attention as the input of c-transformer, which is combined with the transformer and convolutional neural network, to extract the local and global spectral–spatial features. The existing transformer-based methods preserve the location information of pixels through position embedding [52], [54] and dynamically extract the intrinsic relationship through the multihead attention mechanism. However, when the number of labeled samples is small, their spatial distribution diversity is small. It can make these transformer-based methods prone to overfitting the spatial distribution.

C. Contrastive Learning for HSI Classification

Contrastive learning that focuses on fully exploiting unlabeled samples for training has been widely used in HSI classification [55], [56]. The key of contrastive learning-based methods is to construct high-quality positive and negative sample pairs to learn discriminative features [57]. Hou et al. [58] proposed to construct positive and negative samples by adding different random Gaussian noises to unlabeled samples for contrastive learning. Zhao et al. [59] used different transformation operations such as random cropping, horizontal flipping, and Gaussian blur to generate sample pairs. Hu et al. [60] first flipped HSI patches horizontally or vertically, and then randomly removed the spectral information of noncentral pixels in HSI patches to construct different sample pairs. These contrastive learning-based methods elaborately designed different data augmentation strategies to construct positive and negative samples [58], [59]. However, inappropriate data enhancement methods may cause image distortions and interfere with model training. It is challenging to design optimal data enhancement methods for contrastive learning.

III. Proposed Method

A. Overview

Let $\mathcal{X} \in \mathbb{R}^{H\times W\times D}$ represent the HSI, where W, H, and D represent the width, height, and band number, respectively. Semi-supervised HSI classification aims to use a small number of labeled samples $\mathcal{S}_l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$ and a large number of unlabeled samples $\mathcal{S}_u = \{(x_j^u)\}_{j=1}^{N_u}$ in $\mathcal{X}$ to train the network, where $x_i^l \in \mathbb{R}^D$ denotes the ith labeled pixel in $\mathcal{X}$, $y_i^l \in \mathbb{R}^K$ is the one-hot label vector of x_i^l, $x_j^u \in \mathbb{R}^D$ denotes the jth unlabeled pixel, K is the number of categories, N_l is the number of labeled samples, N_u is the number of unlabeled samples, and $N_l \ll N_u$.

This article proposes a PUSL method that aims to fully use unreliable unlabeled samples for semi-supervised HSI classification. The overall framework of PUSL is shown in Fig. 2. First, a PFT is built as the backbone network. Then, weight parameters of PFT are optimized in two stages. In the first stage, only labeled samples are used to optimize PFT to obtain an initial classification network in a supervised manner. The initial classification network is used to split unlabeled samples into reliable and unreliable unlabeled samples. In the second stage, both the labeled samples and unlabeled samples are used to further refine the weight parameters of PFT in a semi-supervised manner. Different from directly discarding unreliable unlabeled samples in the existing pseudolabel-based methods, PUSL uses a contrastive learning paradigm to select unreliable unlabeled samples as negative samples to provide plentiful contrastive information.

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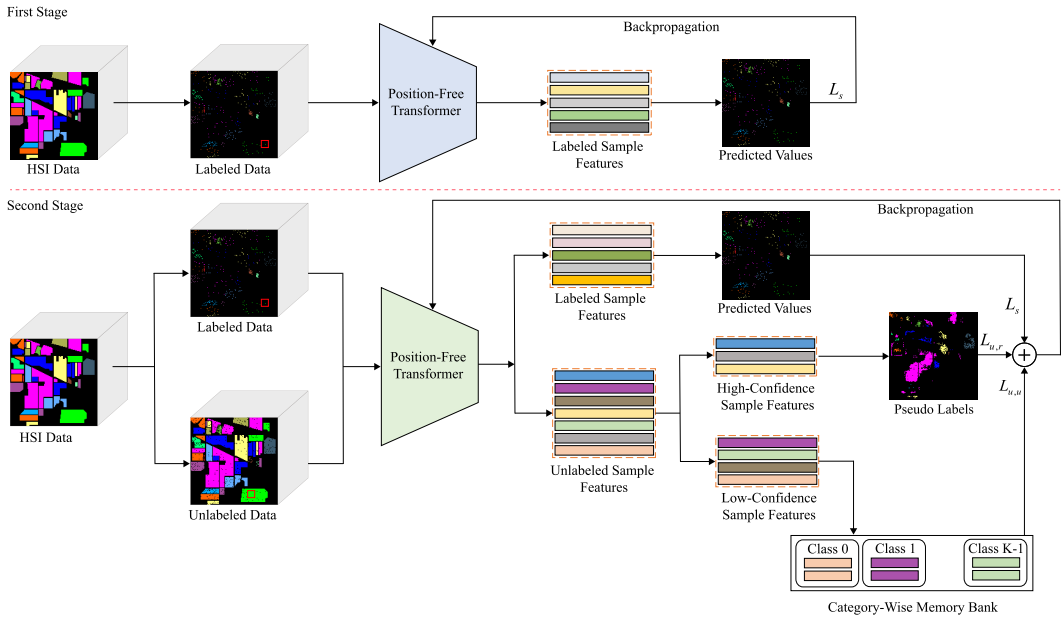

Fig. 2. Detailed structures of PUSL. The training process of PUSL contains two stages. In the first stage, a small number of labeled samples are used to initialize the weight parameters of PFT. In the second stage, both the labeled and unlabeled samples are used to further refine the weight parameters of PFT in a semi-supervised manner. Specially, PUSL uses a contrastive learning paradigm to select unreliable unlabeled samples as negative samples to provide plentiful contrastive information.

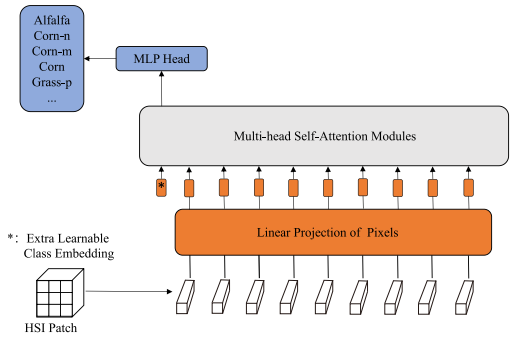

Fig. 3. Structures of PFT.

B. Position-Free Transformer

Both rich spectral bands and spatial distribution of HSI are crucial for the category inference of each pixel [61], [62]. Since categories of adjacent pixels are likely to be the same [63], [64], recent methods usually crop HSI patches that are centered on labeled or unlabeled pixels from the original HSI $\mathcal{X}$ as the input for spectral–spatial feature extraction [7], [65]. For example, an HSI patch $X_i^l \in \mathbb{R}^{h\times w\times D}$ centered on the pixel x_i^l is cropped as the input, where $h \times w$ is the spatial size. Recently, due to the breakthrough progress of ViT in computer vision tasks, several works have attempted to introduce ViT into HSI and construct various transformer-based networks [44], [51], [66]. These works usually follow the original design concept of ViT to encode both the hyperspectral pixels and their corresponding spatial position information into spectral–spatial features. However, the spatial distribution of land covers in HSI is complex due to the changes in imaging perspective and urban construction [67]. Encoding the spatial position information of pixels into the spectral–spatial features may lead to overfitting to the spatial distribution of labeled samples.

In this article, we build a PFT as the backbone for spectral–spatial feature extraction. As shown in Fig. 3, compared with the standard ViT, PFT removes the position embedding module. First, a fully connected layer is used to capture the spectral features from high-dimensional spectral bands of patch in X_i^l. The spectral feature $f_i^l \in \mathbb{R}^M$ of each pixel is used as a pixel token, where M is the feature dimension. Then, hw pixel tokens and a learnable class embedding token $c \in \mathbb{R}^M$ are concatenated as the input $I_i^l \in \mathbb{R}^{(hw+1)\times M}$ of the multihead self-attention modules to extract spectral–spatial feature $F_i^l \in \mathbb{R}^M$ from HSI patch X_i^l

$$I_i^l = \text{Concat}\left[c; f_1^l; \ldots; f_{hw}^l\right] \tag{1}$$

$$F_i^l = \text{MHSA}\left(\text{LN}\left(I_i^l\right)\right) + I_i^l \tag{2}$$

where concat$[\cdot]$ is the concatenation operation, MHSA$(\cdot)$ is the multihead self-attention, and LN$(\cdot)$ is the layer normalization. Finally, the spectral–spatial feature F_i^l is fed to a softmax to obtain the predicted class confidence of each category.

Since PFT with randomly initialized weight parameters cannot effectively divide unlabeled samples into reliable and unreliable unlabeled samples, we adopt a two-stage training scheme. In the first stage, PFT is optimized with the labeled samples and the cross-entropy loss to obtain an initial classification network

$$L_s = \frac{1}{N_l} \sum_{x_i^l \in \mathcal{S}_l} l_{\text{ce}}\left(\text{softmax}\left(F_i^l\right), y_i^l\right) \tag{3}$$

where $l_{\text{ce}}(\cdot)$ is the cross-entropy loss function, and softmax$(\cdot)$ is the softmax classifier. Then, in the second stage, the weights

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TABLE I

CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE IP DATASET

Size	Class	SSAN	SpectralFormer	MDL4OW	DCFSL	CASSN	SSFTT	SS-CNN	SSGPN	3DAES	X-GPN	PUSL
10	1	51.66	87.78	97.83	**100.00**	94.93	95.00	96.74	61.76	97.17	**100.00**	**100.00**
	2	39.25	29.56	58.32	49.11	49.84	48.43	61.10	29.80	44.29	61.40	**80.53**
	3	40.80	32.11	57.57	53.78	55.46	52.90	69.52	23.84	45.73	31.71	**70.12**
	4	53.29	50.84	88.44	89.51	72.86	68.04	87.34	35.11	64.14	**99.65**	93.11
	5	48.87	57.84	81.37	82.07	67.84	**83.99**	82.51	59.66	75.49	79.45	79.85
	6	59.12	55.28	90.97	95.56	86.48	94.54	91.03	96.10	78.71	**99.39**	97.21
	7	68.88	83.33	**100.00**	**100.00**	98.81	98.75	**100.00**	87.50	99.64	**100.00**	**100.00**
	8	62.53	81.88	93.22	95.52	90.73	84.09	98.01	96.14	83.51	**100.00**	98.71
	9	69.67	97.00	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	11.11	**100.00**	**100.00**	**100.00**
	10	45.38	47.65	58.94	73.55	50.58	49.19	60.75	55.42	55.30	74.86	**76.97**
	11	35.72	37.48	59.63	64.72	40.04	45.44	39.45	55.01	48.81	**78.53**	64.09
	12	36.22	32.57	54.10	60.31	55.20	40.84	41.32	36.32	40.46	73.86	**76.81**
	13	68.87	87.74	98.24	**100.00**	97.89	97.95	99.27	91.19	94.44	**100.00**	98.70
	14	61.48	59.07	87.32	93.40	76.07	85.10	74.15	85.63	84.28	**100.00**	91.17
	15	50.60	35.00	82.49	69.47	66.67	64.08	54.15	7.49	63.99	70.43	**83.81**
	16	71.61	88.31	98.39	97.11	99.64	99.73	**100.00**	95.06	98.92	99.28	**100.00**
	OA	65.46±1.32	45.50±1.63	70.15±0.95	72.14±0.56	59.94±0.65	61.59±0.75	63.69±0.28	55.56±1.07	60.20±0.96	77.79±0.79	**79.65±0.72**
	AA	54.00±0.47	60.21±1.91	81.68±0.70	82.76±0.88	75.19±0.78	75.50±0.63	78.46±0.14	57.95±1.05	73.43±0.47	85.53±0.68	**88.19±0.47**
	kappa	61.29±1.39	39.43±1.93	66.39±1.03	68.58±0.57	55.29±0.76	56.95±0.74	59.56±0.28	49.46±1.11	55.44±1.04	74.54±0.87	**77.08±0.83**
20	1	99.28	93.85	89.13	**100.00**	98.55	**100.00**	98.91	95.24	98.70	**100.00**	**100.00**
	2	71.31	42.13	75.56	60.71	63.14	61.89	71.99	53.24	55.48	**86.54**	75.98
	3	75.82	54.57	**84.67**	75.21	72.41	73.42	70.12	47.58	59.49	72.41	83.86
	4	89.87	82.40	97.22	97.51	93.39	83.08	92.19	46.23	75.95	**100.00**	93.67
	5	89.68	79.09	86.71	84.97	84.33	89.45	91.10	72.05	83.15	82.61	**91.51**
	6	91.92	73.99	91.89	99.27	88.49	98.06	97.05	97.02	85.36	**100.00**	95.48
	7	**100.00**	97.50	97.86	**100.00**	98.81	**100.00**	**100.00**	92.31	**100.00**	**100.00**	**100.00**
	8	98.47	92.18	95.15	99.39	92.40	97.82	99.58	97.57	95.31	**100.00**	**100.00**
	9	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
	10	81.86	67.01	81.85	76.24	75.03	74.47	84.57	68.95	66.06	83.72	**93.21**
	11	67.69	56.06	75.18	74.05	60.57	58.28	68.96	66.05	58.91	**82.67**	78.94
	12	72.60	44.84	85.33	84.22	65.99	54.53	86.00	50.00	55.60	47.99	**88.20**
	13	99.92	97.14	92.49	99.89	98.54	99.88	**100.00**	97.22	96.98	**100.00**	**100.00**
	14	85.73	75.65	93.83	93.77	89.75	88.60	90.32	77.82	84.31	**100.00**	95.65
	15	86.92	46.53	89.79	88.85	80.48	87.96	90.67	68.98	77.59	98.36	**98.70**
	16	99.82	96.99	**100.00**	**100.00**	99.28	**100.00**	**100.00**	98.53	98.60	**100.00**	**100.00**
	OA	79.37±0.79	62.31±0.99	83.99±0.70	81.82±0.60	74.74±0.68	74.17±0.66	81.59±0.39	68.08±0.56	69.42±0.60	86.17±0.62	**87.52±0.51**
	AA	88.18±0.63	74.99±0.79	89.79±0.62	89.63±0.57	85.07±0.66	85.46±0.81	90.09±0.56	76.80±1.04	80.72±0.58	90.89±0.53	**93.45±0.32**
	kappa	76.82±0.91	57.63±1.14	81.99±0.81	79.28±0.65	71.59±0.79	70.91±0.74	79.28±0.47	63.60±0.60	65.62±0.68	84.18±0.74	**85.89±0.62**
30	1	**100.00**	96.25	96.30	**100.00**	99.28	98.57	**100.00**	90.91	99.35	**100.00**	**100.00**
	2	75.56	57.68	81.90	81.38	83.87	65.34	75.77	70.28	62.94	82.83	**84.99**
	3	87.23	67.59	90.40	85.50	86.67	78.19	83.07	45.03	68.55	**100.00**	93.76
	4	97.05	85.85	97.85	95.50	90.30	92.69	95.36	90.10	83.12	**100.00**	97.54
	5	90.48	86.40	92.88	87.26	89.99	90.00	97.62	79.24	89.90	**100.00**	93.51
	6	98.77	87.46	91.82	98.31	94.75	95.78	98.49	99.14	90.11	**100.00**	98.13
	7	**100.00**	98.75	98.21	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.64	**100.00**	**100.00**
	8	**100.00**	93.95	96.65	98.70	99.23	98.91	99.90	98.42	97.13	**100.00**	99.90
	9	**100.00**	98.00	96.00	**100.00**	96.67	**100.00**	97.50	**100.00**	99.50	**100.00**	**100.00**
	10	86.52	76.59	83.96	77.47	83.74	78.54	83.02	76.52	70.08	**99.36**	88.12
	11	69.29	62.81	80.35	78.84	67.55	64.36	67.88	67.81	61.36	67.92	**81.71**
	12	88.03	59.22	89.87	87.48	85.33	64.10	80.10	81.72	65.78	**99.82**	92.16
	13	99.51	99.37	96.20	**100.00**	98.70	99.87	**100.00**	97.65	98.88	**100.00**	**100.00**
	14	94.39	87.59	92.89	94.81	94.12	95.16	96.17	84.80	88.31	**100.00**	95.67
	15	82.38	64.89	95.03	91.51	86.27	84.63	90.67	82.91	81.89	**100.00**	96.59
	16	**100.00**	98.41	99.68	**100.00**	99.28	99.65	**100.00**	96.55	99.14	**100.00**	99.64
	OA	84.26±0.51	72.55±0.89	87.59±0.55	87.17±0.47	84.12±0.23	78.19±0.53	83.64±0.35	76.03±0.51	74.35±0.44	89.54±0.44	**90.27±0.39**
	AA	91.82±0.20	82.55±0.72	92.50±0.32	92.30±0.43	90.98±0.61	87.86±0.23	91.60±0.17	85.07±0.70	84.73±0.43	**96.87±0.31**	95.11±0.31
	kappa	82.28±0.57	68.93±0.97	86.02±0.61	85.23±0.52	82.11±0.28	75.27±0.59	81.55±0.37	72.76±0.58	71.15±0.48	88.18±0.51	**88.99±0.46**

of the initial classification network are further optimized using both the labeled and unlabeled samples in a semi-supervised manner. Reliable sample learning and unreliable sample learning are simultaneously exploited in the second training stage. The details of the two training schemes are introduced in Sections III-C and III-D.

C. Reliable Sample Learning

Using the initial classification network PFT obtained in the first stage, we can get the predicted confidence of unlabeled samples in all the categories. First, an unlabeled sample x_j^u passes through PFT to obtain the predicted confidence $p_j^u \in \mathbb{R}^K$. As long as the class confidence in any category is greater than a preset threshold T, this unlabeled sample x_j^u is regarded as a reliable unlabeled sample, and this category is used as the pseudolabel $\hat{p}_j^u \in \mathbb{R}^1$ of this unlabeled sample x_j^u; otherwise, x_j^u is regarded as an unreliable unlabeled sample

$$\hat{p}_j^u = \begin{cases} \text{argmax } p_j^u, & \text{if max } p_j^u > T \\ \text{ignore}, & \text{otherwise.} \end{cases} \tag{4}$$

Let $\mathcal{S}_{u,r}$ denote the set of these reliable unlabeled samples. Then, we refer to reliable unlabeled samples $\mathcal{S}_{u,r}$ together with labeled samples $\mathcal{S}_l$ as reliable samples, which are used to further optimize PFT in a supervised manner. The loss on reliable unlabeled samples $\mathcal{S}_{u,r}$ is

$$L_{u,r} = \frac{1}{|\mathcal{S}_{u,r}|} \sum_{x_j^u \in \mathcal{S}_{u,r}} l_{\text{ce}}\left(\text{softmax}\left(F_j^u\right), \hat{p}_j^u\right) \tag{5}$$

where F_j^u represents the spectral–spatial feature of the jth reliable unlabeled sample x_j^u, and $|\mathcal{S}_{u,r}|$ is the number of reliable unlabeled samples.

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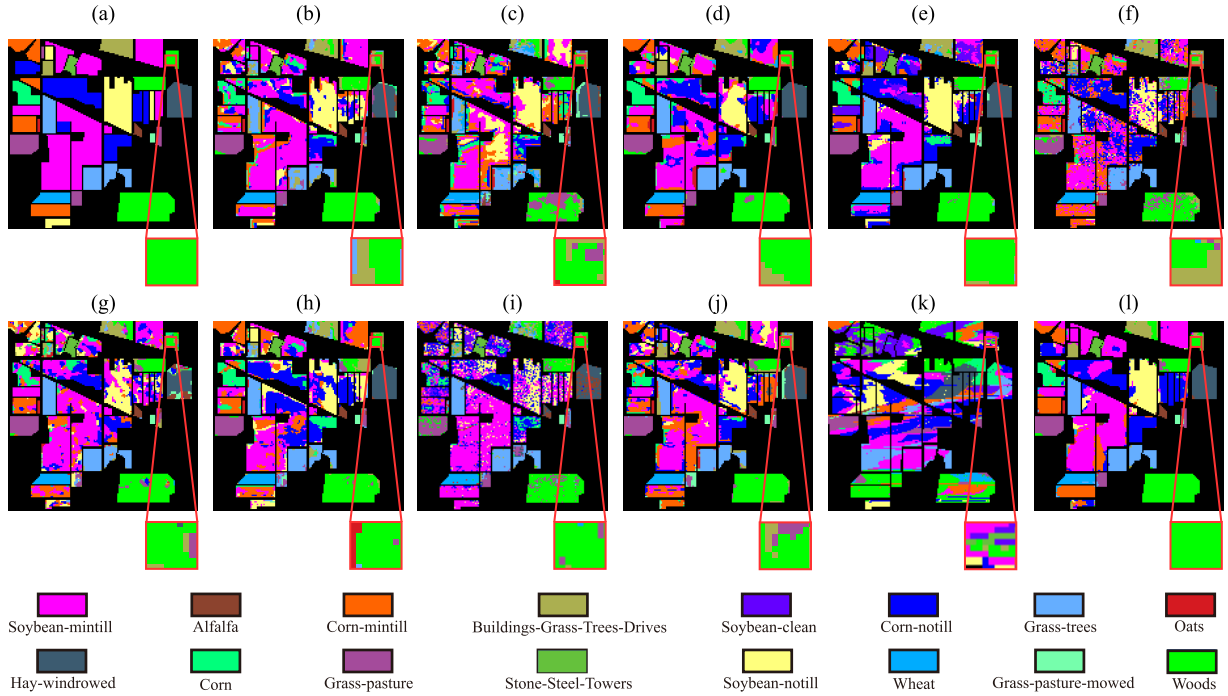

Fig. 4. Classification maps of PUSL and compared methods using ten labeled samples per category. (a) Ground truth. (b) SSAN. (c) SpectralFormer. (d) MDL4OW. (e) DCFSL. (f) CASSN. (g) SSFTT. (h) SS-CNN. (i) SSGPN. (j) 3DAES. (k) X-GPN. (l) PUSL.

Reliable unlabeled samples are used together with labeled samples to optimize PFT. With continuous iterative training (IT), the discrimination of PFT will be gradually enhanced, which is beneficial to select reliable unlabeled samples. Therefore, at each iteration, we use the latest PFT to divide the unlabeled samples into reliable and unreliable unlabeled samples. Specifically, at the tth iteration training, the PFT obtained at the $(t-1)$th iteration is used for division. Through the above operations, the quality of pseudolabel can be gradually improved.

In addition, we leverage a linear adjustment strategy to increase the threshold T as the number of iterations increases for further filtering out low-quality pseudolabel and improving the reliability of pseudolabels. The unlabeled samples whose predicted confidence belongs to the top γ_t are used as reliable unlabeled samples, and the confidence corresponding to the ratio γ_t is just the preset threshold T. Therefore, the threshold T changes as the ratio γ_t changes. In the linear adjustment strategy, the ratio γ_t keeps decreasing as the number of iterations increases

$$\gamma_t = \gamma_0 \cdot \left(1 - \frac{e}{E}\right) \tag{6}$$

where e is the number of current iteration, E is the total number of iterations, and γ_0 is the initial ratio.

D. Unreliable Sample Learning

The existing pseudolabel-based methods usually combine reliable unlabeled samples with labeled samples in a supervised learning manner for training [32], [41], [42], which inevitably makes these methods unable to explore unreliable unlabeled samples. As shown in Fig. 1, according to the predicted confidence, although an unreliable unlabeled sample is confused in the "Alfalfa" and "Oats" categories, it can be determined that this sample does not belong to the "Buildings-g-t-d" category. Based on this phenomenon, we propose to use the contrastive learning paradigm to exploit these unreliable unlabeled samples to improve the discrimination of PFT. Contrastive learning aims to make anchor samples as close as possible to positive samples and as far away from negative samples in the feature space. The settings of anchor samples, positive samples, and negative samples are introduced as follows.

1) Anchor Samples: In the proposed PUSL, both the labeled samples and the reliable unlabeled samples in the current minibatch are selected as anchor samples. Let $\mathcal{A}^k$ represent the set of the kth category anchor samples in the current minibatch. $\mathcal{A}^k$ can be formulated as follows:

$$\mathcal{A}^k = \mathcal{S}_l^k \cup \mathcal{S}_{u,r}^k \tag{7}$$

where $\mathcal{S}_l^k$ and $\mathcal{S}_{u,r}^k$ represent the set of labeled and reliable unlabeled samples belonging to the kth category, respectively.

2) Positive Samples: For the ith anchor sample $x_i^{a,k} \in \mathcal{A}^k$, its corresponding positive sample $x_i^{p,k}$ is the center of all the samples of the same category as this anchor sample in $\mathcal{A}^k$

$$x_i^{p,k} = \frac{1}{|\mathcal{A}^k|} \sum_{x_i^{a,k} \in \mathcal{A}^k} x_i^{a,k} \tag{8}$$

where $|\mathcal{A}^k|$ means the number of samples in $\mathcal{A}^k$. This means that the anchor samples of the same category share the same positive sample, which can encourage samples of the same category to have similar spectral–spatial features.

3) Negative Samples: High-quality negative samples are beneficial to improve the discrimination of PFT. For the anchor sample $x_i^{a,k}$, its negative samples can come from both reliable samples $\mathcal{S}_l \cup \mathcal{S}_{u,r}$ and unreliable unlabeled samples $\mathcal{S}_{u,u}$. Let

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE KSC DATASET

Size	Class	SSAN	SpectralFormer	MDL4OW	DCFSL	CASSN	SSFTT	SS-CNN	SSGPN	3DAES	X-GPN	PUSL
10	1	93.69	94.93	89.38	87.78	86.77	77.54	87.39	94.01	95.58	**100.00**	94.65
	2	85.02	64.25	79.90	88.94	81.21	69.95	80.04	97.00	89.71	**100.00**	97.08
	3	77.58	39.43	88.80	84.27	**91.54**	54.62	86.13	81.30	69.45	67.07	90.59
	4	39.76	27.77	65.61	54.51	50.00	43.64	40.08	52.48	40.32	**90.08**	72.14
	5	73.91	34.24	79.81	75.16	77.85	76.69	76.40	**82.12**	63.98	81.46	77.76
	6	73.19	56.48	86.97	95.02	72.49	61.91	73.80	46.58	76.90	**96.35**	89.69
	7	99.81	95.89	93.17	96.91	92.70	85.79	98.10	73.68	97.52	**100.00**	84.00
	8	71.97	52.38	87.08	**97.64**	77.65	73.51	71.11	85.27	58.56	58.19	75.85
	9	97.19	80.90	93.04	95.90	90.38	83.20	99.23	91.96	89.88	**100.00**	98.17
	10	96.19	79.70	81.60	99.24	85.56	74.91	97.28	92.13	85.89	**100.00**	97.38
	11	95.70	97.24	95.78	96.35	93.00	81.50	99.52	89.73	97.66	**100.00**	**100.00**
	12	88.95	87.06	78.76	90.71	73.76	49.29	92.54	84.79	88.95	72.82	**94.19**
	13	**100.00**	**100.00**	93.82	**100.00**	91.87	98.63	99.95	99.78	**100.00**	**100.00**	99.87
	OA	88.07±0.52	78.42±0.72	87.29±0.65	91.87±0.75	83.85±0.57	75.43±0.67	88.28±0.29	87.50±0.51	85.52±0.64	91.12±0.51	**92.84±0.43**
	AA	84.07±0.79	70.02±0.93	85.67±0.55	89.42±0.61	81.91±0.55	71.63±0.78	84.74±0.18	82.37±0.45	81.11±0.69	89.69±0.33	**90.11±0.46**
	kappa	86.72±0.58	75.96±0.79	85.81±0.73	90.96±0.81	82.07±0.66	72.63±0.63	86.98±0.33	86.06±0.57	83.89±0.73	90.08±0.57	**92.03±0.51**
20	1	97.87	96.71	94.55	89.48	94.74	88.48	94.68	97.71	97.56	**100.00**	98.82
	2	95.64	94.57	90.26	95.71	92.04	70.15	92.39	95.52	93.29	**100.00**	95.23
	3	90.86	73.14	93.95	87.80	**96.09**	63.92	87.89	84.32	83.05	92.07	95.43
	4	68.73	38.75	**83.53**	74.38	81.75	55.02	51.39	74.57	51.67	52.07	81.55
	5	85.59	62.48	84.16	**90.73**	79.09	90.29	79.19	81.56	79.75	81.65	87.58
	6	86.99	80.00	94.83	97.72	87.92	81.38	87.99	82.78	85.24	**99.09**	92.40
	7	**100.00**	96.12	96.51	87.37	98.73	96.49	97.14	96.47	**100.00**	**100.00**	99.62
	8	89.74	82.36	96.02	**100.00**	90.95	66.04	92.92	85.89	82.27	98.78	98.12
	9	**100.00**	93.54	97.24	**100.00**	98.14	91.22	99.90	98.00	96.71	**100.00**	99.52
	10	96.19	90.68	91.96	**100.00**	94.88	78.56	98.89	94.79	86.61	98.98	96.21
	11	99.86	99.12	96.90	93.89	99.60	87.88	99.88	99.25	99.38	**100.00**	**100.00**
	12	94.43	93.29	91.48	**98.17**	92.58	65.75	94.93	84.06	90.00	95.18	89.72
	13	**100.00**	**100.00**	99.08	**100.00**	98.99	98.04	**100.00**	99.67	**100.00**	**100.00**	**100.00**
	OA	94.81±0.46	89.57±0.51	94.41±0.56	95.16±0.61	94.33±0.44	81.93±0.52	93.51±0.20	92.75±0.28	91.04±0.55	96.11±0.43	**96.27±0.32**
	AA	92.76±0.63	84.67±0.81	93.11±0.52	93.48±0.58	92.73±0.53	79.48±0.42	90.55±0.22	90.35±0.41	88.12±0.61	93.68±0.31	**94.94±0.36**
	kappa	94.23±0.51	88.35±0.65	93.76±0.62	94.61±0.67	93.69±0.51	79.86±0.57	92.78±0.23	91.90±0.30	90.03±0.59	95.66±0.47	**95.85±0.33**
30	1	96.85	98.04	94.57	98.27	97.24	96.38	98.62	99.73	97.71	**100.00**	99.95
	2	99.59	94.88	94.86	96.15	96.02	80.83	91.36	98.12	96.58	**100.00**	97.53
	3	96.48	75.53	94.08	91.68	96.22	77.93	88.28	87.17	86.99	**99.78**	95.82
	4	84.52	53.70	**89.48**	73.51	79.76	47.80	63.29	76.58	63.89	79.05	88.53
	5	**93.17**	75.35	91.51	90.84	90.68	79.83	80.43	88.55	84.29	89.69	92.05
	6	90.39	85.68	94.98	98.85	87.92	82.23	94.98	68.84	87.86	**100.00**	95.20
	7	**100.00**	99.33	96.51	94.79	99.05	99.10	95.71	**100.00**	99.90	**100.00**	99.62
	8	96.29	90.67	95.82	**100.00**	96.13	78.28	97.68	95.01	92.55	**100.00**	98.42
	9	98.85	96.51	98.01	**100.00**	99.04	84.17	**100.00**	98.57	96.31	98.98	98.73
	10	**100.00**	96.31	96.08	99.88	98.60	94.24	98.89	95.99	91.61	99.33	98.56
	11	**100.00**	98.84	95.54	98.66	99.36	95.66	99.88	99.74	99.47	86.89	99.98
	12	93.44	93.30	94.76	**98.46**	94.70	74.69	96.22	91.97	91.93	98.10	96.14
	13	**100.00**	**100.00**	99.77	**100.00**	99.68	95.17	**100.00**	98.33	**100.00**	**100.00**	99.95
	OA	96.91±0.30	92.76±0.41	95.86±0.48	97.23±0.46	96.26±0.31	86.27±0.41	95.48±0.22	94.61±0.27	93.64±0.40	97.34±0.28	**97.90±0.28**
	AA	96.12±0.30	89.09±0.65	95.07±0.46	95.47±0.32	94.95±0.23	83.56±0.48	92.72±0.23	92.20±0.31	91.47±0.36	96.29±0.21	**96.96±0.26**
	kappa	96.56±0.34	91.90±0.46	95.38±0.52	96.91±0.51	95.84±0.35	84.65±0.45	94.97±0.25	93.96±0.30	92.93±0.44	97.03±0.32	**97.66±0.31**

$\mathcal{N}^k$ represent the set of negative samples of the kth category. $\mathcal{N}^k$ can be formulated as follows:

$$\mathcal{N}^k = \mathcal{N}_r^k \cup \mathcal{N}_{u,u}^k \tag{9}$$

where $\mathcal{N}_r^k$ and $\mathcal{N}_{u,u}^k$ represent the set of negative samples of the kth category from $\mathcal{S}_l \cup \mathcal{S}_{u,r}$ and $\mathcal{S}_{u,u}$, respectively.

In reliable samples $\mathcal{S}_l \cup \mathcal{S}_{u,r}$, negative samples should have two characteristics: the negative sample should not belong to the kth category and should be indistinguishable from the anchor sample $x_i^{a,k}$. In other words, negative sample also has a large confidence in the kth category

$$\mathcal{N}_r^k = \left\{ x_i^r \middle| \hat{p}_i^r \neq k, 0 \leq O\left(p_i^r, k\right) < r_l \right\} \tag{10}$$

where $x_i^r \in \mathcal{S}_l \cup \mathcal{S}_{u,r}$, $\hat{p}_i^r$ is the predicted label of the reliable sample x_i^r, and $p_i^r \in \mathbb{R}^K$ is the predicted confidence vector of x_i^r in all the categories, $O(p_i^r, k)$ is sorted in the descending sort of the class confidence of reliable samples. The value of $O(p_i^r, k)$ is the class confidence rank of p_i^r in the kth category. A small value of $O(p_i^r, k)$ indicates a large confidence in the category k. r_l is the low ranking threshold, and r_l is set to 3.

In unreliable unlabeled samples $\mathcal{S}_{u,u}$, negative samples should have a greater probability of not belonging to the kth category

$$\mathcal{N}_{u,u}^k = \left\{ x_i^{u,u} \middle| r_l \leq O\left(p_i^{u,u}, k\right) < r_h \right\} \tag{11}$$

where $x_i^{u,u} \in \mathcal{S}_{u,u}$, $p_i^{u,u} \in \mathbb{R}^K$ is the predicted confidence vector of the unreliable unlabeled sample $x_i^{u,u}$ in all the categories, $O(p_i^{u,u}, k)$ is the class confidence ranking of $x_i^{u,u}$ in the kth category, r_h is the threshold of high rank, and r_h is set to the number of classes K.

4) Categorywise Memory Bank: In a minibatch of the training phase, the number of samples of different categories is different. If the model selects the corresponding negative samples for all the categories from only the current minibatch, this may lead to the imbalance of the number of candidate negative samples for different categories [68], [69], [70]. To maintain a stable number of negative samples, the proposed PUSL builds a categorywise memory bank to allocate a fixed-size queue for each category to store candidate negative samples. In practice, the constructed categorywise memory bank does not directly store these candidate negative samples, but stores spectral–spatial features of these candidate negative samples to avoid redundant feature extraction operations. PUSL dynamically updates the categorywise memory bank in

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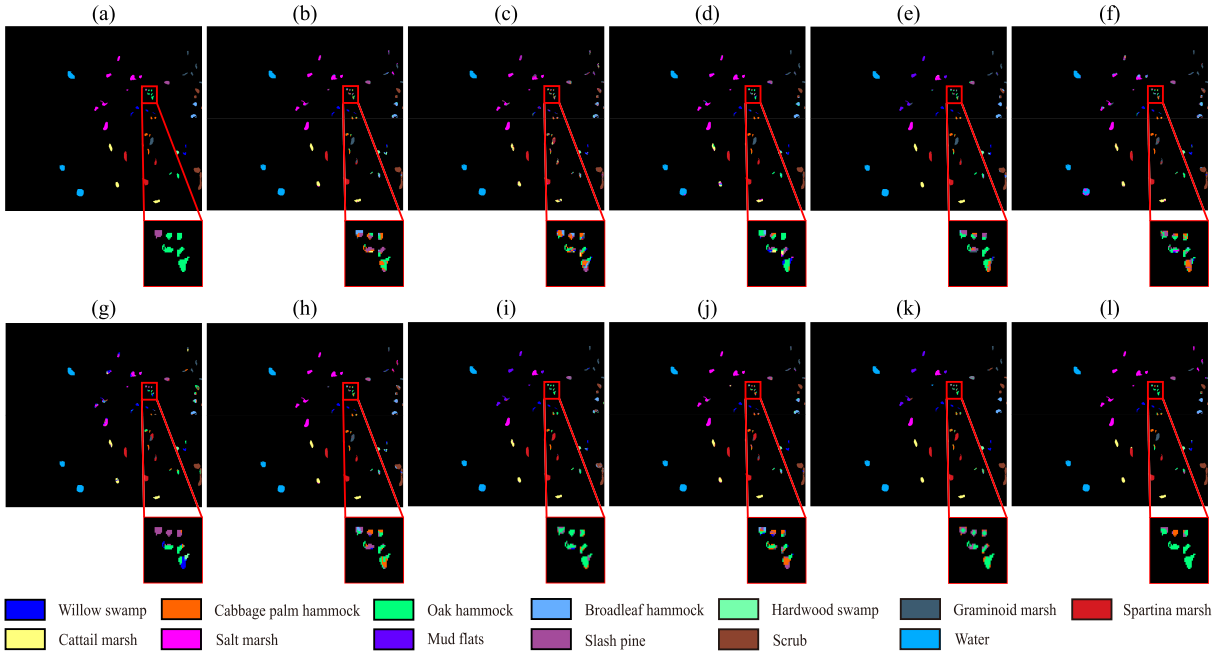

Fig. 5. Classification maps of PUSL and compared methods using ten labeled samples per category. (a) Ground truth. (b) SSAN. (c) SpectralFormer. (d) MDL4OW. (e) DCFSL. (f) CASSN. (g) SSFTT. (h) SS-CNN. (i) SSGPN. (j) 3DAES. (k) X-GPN. (l) PUSL.

each iteration. In the tth iteration, the spectral–spatial features of new candidate negative samples will replace the old in a first-in-first-out manner for each category.

Through the above operations, we can obtain anchor samples $x_i^{a,k} \in \mathcal{A}^k$, positive sample $x_i^{p,k}$, and negative samples $x_i^{n,k} \in \mathcal{N}^k$ of each category. Let $F_i^{a,k}$, $F_i^{p,k}$, and $F_i^{n,k} \in \mathbb{R}^M$ represent the spectral–spatial features of $x_i^{a,k}$, $x_i^{p,k}$, and $x_i^{n,k}$, respectively. The contrastive loss $L_{u,u}$ can be calculated as

$$L_{u,u} = -\frac{1}{K|\mathcal{A}^k|}\sum_{k=1}^{K}\sum_{i=1}^{|\mathcal{A}^k|}\log \times\left[\frac{e^{\left\langle F_i^{a,k},F_i^{p,k}\right\rangle/\tau}}{e^{\left\langle F_i^{a,k},F_i^{p,k}\right\rangle/\tau}+\sum_{j=1}^{Q^k}e^{\left\langle F_i^{a,k},F_j^{n,k}\right\rangle/\tau}}\right] \tag{12}$$

where K is the number of categories, $|\mathcal{A}^k|$ is the number of anchor samples, Q^k is the number of negative samples for each category in the categorywise memory bank, $\langle\cdot,\cdot\rangle$ is the cosine similarity, τ is a temperature hyperparameter, and τ is set to 0.5.

The overall loss function of the proposed PUSL in the second stage is as follows:

$$L = L_s + \lambda_{u,r}L_{u,r} + \lambda_{u,u}L_{u,u} \tag{13}$$

where $\lambda_{u,r}$ and $\lambda_{u,u}$ are the hyperparameters to control the balance of different terms.

IV. Experiments

A. Evaluation Measures and Datasets

To quantitatively evaluate the performance of PUSL, we use overall accuracy (OA), average accuracy (AA), kappa, and accuracy for each class as evaluation measures [48]. The Indian Pines (IP), Kennedy Space Center (KSC), and Houston2013 datasets are used to conduct extensive experiments.

1) IP Dataset: This dataset includes 145 × 145 pixels. The original dataset includes 224 bands, and the remaining 200 bands are used for HSI classification after removing 24 interfered bands. The central wavelength of the IP dataset is 400–2500 nm, and the spatial resolution is 20 m. The IP contains 16 categories of land covers.

2) KSC Dataset: The KSC dataset was captured at KSC above Florida through the AVIRIS. The original dataset includes 224 bands, and the remaining 176 bands are used for experiments after removing 48 interfered bands. The central wavelength of the KSC dataset is 400–2500 nm, and the spatial resolution size is 18 m. The image size of KSC is 512 × 641. The KSC dataset contains 13 kinds of land covers.

3) Houston2013 Dataset: The Houston2013 dataset was captured over the University of Houston by the ITRES CasI-1500 sensor. The image size is 349 × 1905. This dataset contains 144 bands in the spectral range from 364 to 1046 nm and contains 15 categories of land covers.

B. Experimental Settings

To evaluate the performance of PUSL under different numbers of labeled samples on different datasets, 10, 20, and 30 samples are randomly selected from each class as labeled samples, and the remaining samples are used as unlabeled samples, respectively. It is particularly emphasized that due to the limited number of the "Oats" and "Grass-pasture-mowed" categories on the IP dataset, 10 and 20 samples are selected as labeled samples in experiments of 20 and 30 samples per category, respectively. PUSL adopts the Adam optimizer and sets the batch size to 256 for training. The learning rate of PUSL is set to 0.001, 0.0001, and 0.0001 on the IP, KSC, and Houston2013 datasets, respectively. PFT contains four multihead self-attention modules, and each module has four heads. We set the dimension M of spectral–spatial feature

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TABLE III

CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

Size	Class	SSAN	SpectralFormer	MDL4OW	DCFSL	CASSN	SSFTT	SS-CNN	SSGPN	3DAES	X-GPN	PUSL
10	1	92.60	94.90	83.93	**95.73**	94.22	90.81	89.41	88.80	90.67	94.74	87.42
	2	94.18	86.24	85.29	96.46	90.59	88.19	96.29	96.70	88.71	**96.78**	94.21
	3	98.88	91.63	92.61	96.07	95.89	99.70	99.14	99.71	94.48	99.96	**100.00**
	4	94.00	87.79	89.63	92.06	93.41	93.23	92.64	94.65	88.98	**98.00**	95.00
	5	97.76	95.27	87.18	98.62	99.28	99.95	98.83	97.65	96.80	**100.00**	99.75
	6	88.18	72.92	86.00	89.52	97.33	94.82	86.31	**99.68**	86.06	95.16	95.23
	7	67.32	61.61	68.32	62.32	**76.97**	74.12	45.78	70.03	71.29	58.29	74.92
	8	64.37	60.81	58.43	67.10	58.76	63.54	57.44	55.83	55.65	60.96	**79.45**
	9	67.36	65.52	74.04	**74.15**	71.70	72.28	52.40	62.80	71.78	45.69	68.96
	10	73.69	66.20	75.81	71.65	73.32	**89.54**	75.02	57.44	64.16	70.35	81.44
	11	68.63	63.22	70.00	74.78	73.63	**89.54**	81.62	72.33	60.71	54.71	73.05
	12	82.66	69.83	70.40	**92.40**	81.91	75.35	65.77	49.14	68.80	45.18	85.64
	13	85.50	35.47	89.98	93.25	70.22	94.31	71.54	65.80	66.91	89.22	**94.93**
	14	99.91	94.31	89.64	98.80	98.75	97.71	85.86	97.61	95.05	**100.00**	96.40
	15	99.45	92.68	**100.00**	98.62	99.70	99.69	97.95	99.08	91.59	**100.00**	99.83
	OA	82.85±0.96	75.94±0.75	79.12±0.94	84.78±0.77	83.36±0.29	86.01±0.63	77.97±0.46	77.73±0.68	77.85±0.86	76.72±0.61	**86.35±0.50**
	AA	84.97±0.71	75.89±0.63	81.42±0.68	86.77±0.50	85.05±0.36	88.19±0.56	79.73±0.41	80.48±0.65	79.44±0.75	80.60±0.56	**88.42±0.60**
	kappa	81.47±1.05	74.01±0.91	77.52±1.00	83.56±0.89	82.02±0.32	84.88±0.68	76.17±0.47	75.93±0.71	76.07±0.93	74.84±0.67	**85.26±0.65**
20	1	95.12	97.50	90.61	94.39	96.24	91.84	96.68	96.17	95.80	95.37	**97.75**
	2	86.60	93.77	89.83	96.27	94.90	86.46	95.45	94.56	91.58	**96.76**	95.93
	3	99.86	95.11	96.80	98.97	97.80	98.33	98.28	99.11	96.44	**100.00**	99.48
	4	92.44	93.24	95.73	96.73	94.02	**99.78**	93.77	95.90	89.63	95.79	93.41
	5	**100.00**	97.86	94.47	96.24	98.87	99.62	99.48	99.43	98.33	99.98	98.38
	6	88.00	80.46	**98.77**	89.51	96.92	95.91	85.69	96.36	91.45	95.98	96.65
	7	83.91	77.41	85.06	88.30	81.83	84.22	82.93	**89.64**	81.32	88.10	86.40
	8	71.70	74.80	78.95	74.84	75.59	75.86	70.86	67.65	71.37	69.57	**86.13**
	9	**82.43**	75.15	81.30	75.65	81.18	69.68	64.54	58.50	77.66	67.39	77.84
	10	**90.06**	87.31	87.50	87.90	85.71	87.44	89.32	84.30	82.21	78.87	85.31
	11	85.51	73.99	80.78	83.87	87.23	**93.29**	80.85	83.33	71.21	85.51	87.94
	12	84.83	83.85	86.39	**96.87**	86.86	82.14	88.73	74.71	82.86	74.28	93.69
	13	92.75	50.31	**95.38**	92.87	76.83	94.85	85.71	53.81	73.01	92.76	95.05
	14	99.07	96.71	98.09	99.26	**100.00**	97.06	98.48	99.26	96.85	**100.00**	99.74
	15	99.70	97.63	99.95	98.12	99.55	**100.00**	98.48	98.59	96.14	**100.00**	99.88
	OA	88.91±0.70	85.59±0.87	88.91±0.74	90.35±0.88	89.35±0.44	88.80±1.02	87.66±0.34	85.45±0.51	85.47±0.48	87.38±0.46	**91.68±0.38**
	AA	90.13±0.51	85.01±0.83	90.64±0.54	91.32±0.83	90.24±0.48	90.43±0.88	88.62±0.62	86.09±0.61	86.39±0.37	89.36±0.49	**92.91±0.46**
	kappa	88.01±0.75	84.41±0.98	88.03±0.80	89.57±1.03	88.48±0.49	87.89±1.10	86.66±0.37	84.26±0.65	84.29±0.51	86.35±0.55	**91.01±0.50**
30	1	99.20	96.04	93.47	95.21	95.71	96.45	**99.76**	86.20	95.00	97.93	95.45
	2	89.95	94.35	94.32	98.14	93.14	88.02	98.13	96.31	93.44	93.93	**98.16**
	3	97.42	98.43	96.94	99.85	99.71	97.50	99.57	99.25	97.96	**100.00**	99.99
	4	94.37	95.79	92.59	95.02	95.79	92.59	93.61	**99.59**	93.71	95.68	91.95
	5	97.42	98.03	94.57	97.46	98.95	98.72	99.24	99.17	97.94	**100.00**	98.27
	6	98.77	89.05	94.67	90.16	95.59	97.74	94.00	**100.00**	93.69	99.15	99.57
	7	**92.11**	82.97	84.53	89.82	90.88	85.19	80.01	86.22	82.45	90.91	87.70
	8	67.68	78.12	**85.79**	75.25	79.58	83.35	71.14	61.49	77.25	71.71	82.99
	9	82.59	81.73	82.61	79.55	**86.55**	74.64	85.62	72.33	81.10	81.40	85.78
	10	89.49	82.45	90.94	90.64	90.49	90.25	87.69	82.65	81.86	**91.73**	90.43
	11	86.07	81.20	90.03	85.35	86.64	91.71	91.01	90.01	78.20	**92.32**	91.42
	12	85.16	87.57	90.73	95.38	87.62	**96.12**	78.43	70.56	86.18	75.54	89.67
	13	92.96	57.22	97.73	95.77	82.16	**99.75**	89.45	83.45	78.38	96.53	96.82
	14	**100.00**	98.82	95.64	99.26	**100.00**	99.44	99.65	99.24	97.57	**100.00**	99.91
	15	**100.00**	98.57	99.95	97.81	99.60	**100.00**	99.55	99.20	96.92	**100.00**	**100.00**
	OA	90.04±0.56	88.19±0.52	91.22±0.49	91.38±0.43	91.48±0.41	91.21±0.49	89.94±0.16	86.46±0.47	87.89±0.34	90.82±0.39	**92.58±0.36**
	AA	91.55±0.42	88.02±0.41	92.30±0.48	92.31±0.24	92.16±0.42	92.76±0.46	91.12±0.11	88.38±0.50	88.78±0.33	92.46±0.27	**93.87±0.37**
	kappa	89.23+0.60	87.22±0.61	90.52±0.52	90.68±0.50	90.79±0.45	90.50±0.55	89.12±0.20	85.36±0.50	86.91±0.37	90.07±0.46	**91.98±0.44**

F_i^l to 128 in experiments. In the categorywise memory bank, we set the size of the queue to 256 for each category to store candidate negative samples. All the experiments are implemented with the PyTorch framework on a workstation equipped with GeForce RTX 3090. To fairly compare the performance of PUSL with other methods, all the methods use the same labeled training samples and set the same patch size 7×7. The experiments of each method are repeated ten times, and the average values are reported as the final results.

C. Compared Methods

To demonstrate the performance of PUSL, ten state-of-the-art methods are used as compared methods, including spectral-spatial attention network (SSAN) [48], SpectralFormer [66], multitask deep learning method for open world (MDL4OW) [71], deep cross-domain few-shot learning (DCFSL) [72], cross-attention spectral-spatial network (CASSN) [65], spectral-spatial feature tokenization transformer (SSFTT) [52], semi-supervised convolutional neural network (SS-CNN) [25], semi-supervised graph prototypical network (SSGPN) [73], 3D autoencoder and siamese network (3DAES) [24], and cross-scale graph prototypical network (X-GPN) [67]. Details of the compared methods are introduced as follows.

1) *SSAN [48]:* SSAN extracts features through self-attention modules and 3-D convolution operations.
2) *SpectralFormer [66]:* SpectralFormer adopts a ViT to mine the correlations among different spectrums.
3) *MDL4OW [71]:* MDL4OW is a multitask reconstruction and classification network for HSI classification.
4) *DCFSL [72]:* DCFSL overcomes the domain shift through a conditional adversarial domain adaptation strategy.
5) *CASSN [65]:* First, local and global band correlations are extracted by spectral attention, and then spatial information of HSI is aggregated by cross spatial attention.
6) *SSFTT [52]:* SSFTT uses the shallow features extracted by a convolutional neural network as the input of a transformer to further obtain the spectral–spatial information of HSI patches.
7) *SS-CNN [25]:* SS-CNN builds an encoder–decoder network with a skip connection. Then, classification and reconstruction losses are used to optimize the network parameters.
8) *SSGPN [73]:* SSGPN is a graph-based method that uses a distance-based cross-entropy function and a temporal entropy-based regularizer to improve the model discrimination.

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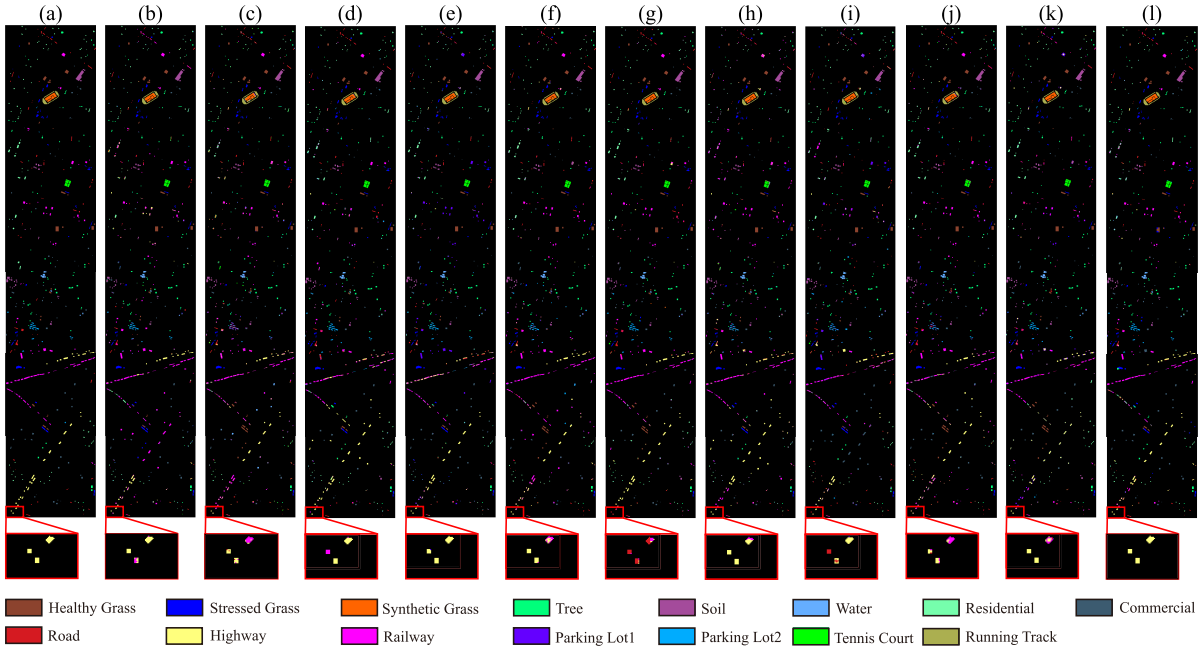

Fig. 6. Classification maps of PUSL and compared methods using ten labeled samples per category. (a) Ground truth. (b) SSAN. (c) SpectralFormer. (d) MDL4OW. (e) DCFSL. (f) CASSN. (g) SSFTT. (h) SS-CNN. (i) SSGPN. (j) 3DAES. (k) X-GPN. (l) PUSL.

TABLE IV

IMPACT OF RELIABLE AND UNRELIABLE UNLABELED SAMPLES

Database	IP			KSC			Houston2013		
Number	10	20	30	10	20	30	10	20	30
PFT	68.45	83.04	85.46	88.26	92.84	94.97	84.05	87.85	89.31
PFT+RU	77.70	85.87	88.37	90.62	94.55	96.43	85.28	90.01	91.44
PUSL	**79.65**	**87.52**	**90.27**	**92.84**	**96.27**	**97.90**	**86.35**	**91.68**	**92.58**

9) *3DAES [24]:* 3DAES uses an autoencoder to extract the spectral–spatial features from a large number of unlabeled samples, and then optimizes the Siamese network and classifier using constructed sample pairs.
10) *X-GPN [67]:* X-GPN represents the topological information of HSI through a multiscale graph structure to obtain rich spectral–spatial information.

D. Experimental Results

1) IP Dataset: First, the IP dataset is used to evaluate the performance of PUSL and compared methods. To comprehensively evaluate the performance of different methods, we set the number of labeled samples in each category as 10, 20, and 30, respectively. The experimental results are shown in Table I. SSAN and CASSN use an attention-based network to extract the spectral–spatial features of HSI patches. The OAs of SSAN and CASSN on ten labeled samples per category for training are 65.46% and 59.94%, respectively. When the number of labeled samples is small, most of the compared methods are prone to overfitting. The OAs of SpectralFormer, CASSN, SSFTT, SSGPN, and 3DAES using ten labeled samples per category for training are 45.50%, 59.94%, 61.59%, 55.56%, and 60.20%, respectively. The OA of DCFSL in the experiment using ten labeled samples per category for training is 72.14%. The proposed PUSL uses contrastive learning to fully exploit unreliable unlabeled samples for training. PUSL achieves the best performance in terms of OA, AA, and kappa.

Fig. 4 shows visual classification maps of PUSL and compared methods on the IP dataset. The samples at the edges of land covers have complex spatial distributions, which lead to poor performance of SSAN, SpectralFormer, CASSN, and SSFTT. The classification maps of SS-CNN and 3DAES are also not smooth because the features extracted by an encoder–decoder are not discriminative enough for accurate HSI classification. SSGPN fuses the spectral features of all the pixels in a region and treats the region as a node. This method may introduce interfering pixels whose categories are different from the central pixel, which leads to easy misclassification on ten labeled samples per category for training. The classification map of PUSL is relatively smooth in the edge regions.

2) KSC Dataset: The experimental results of PUSL and compared methods are reported in Table II. In terms of OA, AA, and kappa, PUSL achieves the best classification performance under different numbers of labeled samples. SpectralFormer and SSFTT use a transformer-based network to extract spectral–spatial information of HSI patches. These methods are prone to overfitting with a small number of labeled samples. In experiments using ten labeled samples per category for training, the OAs of SpectralFormer and SSFTT are 78.42% and 75.43%, respectively. MDL4OW uses an encoder–encoder to reconstruct the input features, which is difficult to effectively distinguish the label information of the samples in a small number of labeled samples. 3DAES optimizes the Siamese network using constructed sample pairs to obtain better classification performance. X-GPN constructs a multigraph structure to represent the fine-grained

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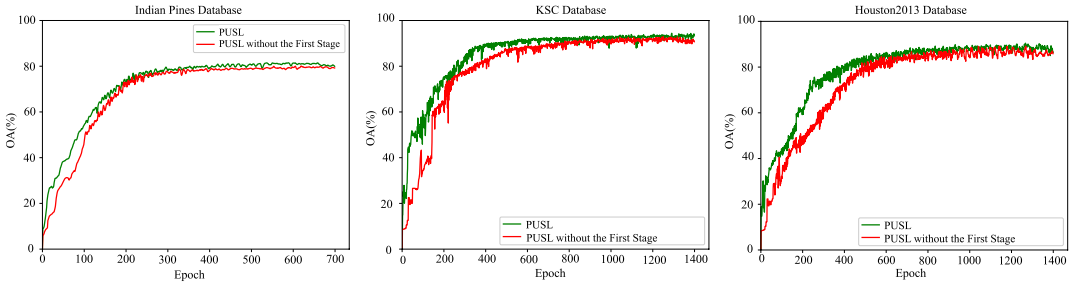

Fig. 7. OAs (%) of PUSL on the test set at different training epochs in the experiment using ten labeled samples per category for training.

TABLE V
IMPACT OF CONTINUOUS IT

Database	IP	KSC	Houston2013
PUSL without IT	77.59	90.10	82.27
PUSL	**79.65**	**92.84**	**86.35**

TABLE VI
IMPACT OF PE ON THE PERFORMANCE OF PFT AND PUSL

Database	Methods	10	20	30
IP	PFT with PE	52.21	73.86	77.70
	PFT	68.45	83.04	85.46
	PUSL with PE	59.32	78.86	81.39
	PUSL	**79.65**	**87.52**	**90.27**
KSC	PFT with PE	69.89	77.54	83.90
	PFT	88.26	92.84	94.97
	PUSL with PE	76.98	82.88	87.58
	PUSL	**92.84**	**96.27**	**97.90**
Houton2013	PFT with PE	78.65	82.01	86.45
	PFT	84.05	87.85	89.31
	PUSL with PE	80.24	85.31	88.47
	PUSL	**86.35**	**91.68**	**92.58**

spectral–spatial features of the HSI. Compared with 3DAES, the OA of X-GPN on ten labeled samples per category for training is improved by 5.60%. Due to the strong feature extraction ability of PFT and the full utilization of unreliable unlabeled samples, PUSL outperforms the compared methods in terms of OA, AA, and kappa.

Fig. 5 shows visual classification maps of different methods on the KSC dataset. The spatial distributions of different kinds of land covers on the KSC dataset are scattered. As shown in Fig. 5, most of the compared methods have poor performance in small regions. The feature extraction ability of SSAN, SpectralFormer, and CASSN is limited in a small number of labeled samples. DCFSL and 3DAES are difficult to accurately fine-tune the network parameters and are prone to overfitting with a small number of labeled samples. The proposed PUSL achieves smooth classification map in small regions. For example, PUSL achieves superior performance in the "Salt marsh" and "Mud flats" categories.

3) Houston2013 Dataset: The experimental results of different methods are reported in Table III. SpectralFormer tends to overfit and achieves poor classification performance in a small number of labeled samples. Compared with SpectralFormer, SSFTT adds rich spatial information of adjacent pixels, and the OA is improved by 10.07% in the experiment using ten labeled samples per category for training. SS-CNN and 3DAES focus on reconstructing the original input, and the OAs of SS-CNN and 3DAES using ten labeled samples per category for training are 77.97% and 77.85%, respectively. SSGPN and X-GPN regard a small area as a node, and then represent the spatial information of the HSI through an adjacency matrix. Due to discrete distributions of samples on the Houston2013 dataset, SSGPN and X-GPN can also introduce more interfering pixels whose categories are different from the center pixels. The OAs of SSGPN and X-GPN using ten labeled samples per category for training are 77.73% and 76.72%, respectively. The proposed PUSL also achieves the best performance in terms of OA, AA, and kappa in experiments using 10, 20, and 30 labeled samples per category.

Fig. 6 displays visual classification maps of different methods on the Houston2013 dataset. Discrete distributions of samples pose a great challenge to different methods. It can be observed that the compared methods have obvious misclassification in small regions. SpectralFormer and SSFTT are easy to cause category misjudgment under a small number of labeled samples. SSAN and CASSN use an attention mechanism to the extract spectral–spatial features of samples, which are prone to misclassification with a small number of labeled samples. The proposed PUSL still achieves superior classification performance.

E. Ablation Study

1) Impact of Reliable and Unreliable Unlabeled Samples: Previous pseudolabel-based semi-supervised methods usually focus on exploiting reliable unlabeled samples, leaving unreliable unlabeled samples underused. In this article, we propose to use a contrastive learning paradigm to exploit unreliable unlabeled samples as negative samples for semi-supervised training. To verify the effectiveness of reliable and unreliable unlabeled samples, we conduct several experiments for ablation analysis. The results are reported in Table IV. PFT means that only labeled samples are used for training. PFT + RU means that labeled samples and reliable unlabeled samples are used for training. PUSL is the proposed methods that uses both the labeled and unlabeled samples for training. The experimental results show that using both the reliable unlabeled samples and unreliable unlabeled samples can effectively improve the classification performance.

2) Impact of the First Stage: Several experiments are conducted to explore the impact of the first stage training strategy on the performance of PUSL. Fig. 7 records the accuracy curves of PUSL and PUSL without the first stage training strategy in the experiment using ten labeled samples

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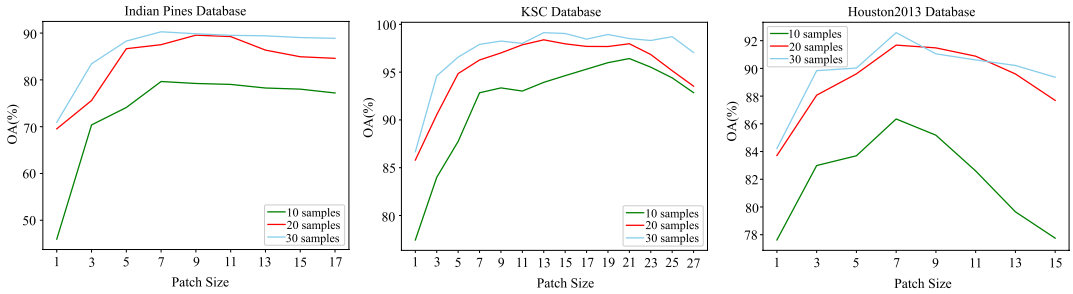

Fig. 8. OAs (%) of the proposed PUSL under different spatial sizes of HSI patches.

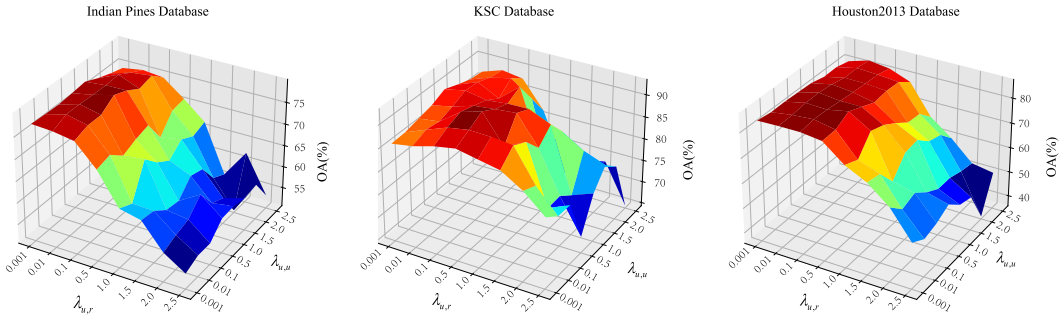

Fig. 9. OAs (%) of PUSL under different hyperparameters $\lambda_{u,r}$ and $\lambda_{u,u}$ using ten labeled samples per category for training.

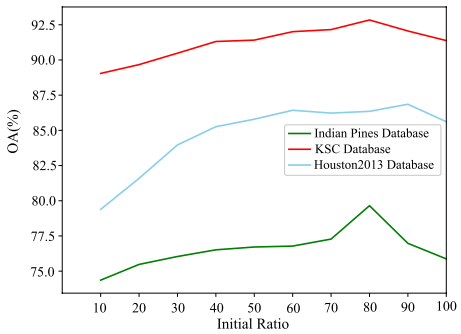

Fig. 10. OAs (%) of PUSL under different initial ratios using ten labeled samples per category.

per category for training on the test set during the training process. In the first stage, the proposed PUSL only uses a small number of labeled samples to initialize the weight parameters of PFT in a supervised training manner, which is beneficial to extract spectral–spatial features from unlabeled samples in the second stage training process. As shown in Fig. 7, the first stage training strategy of PUSL can speed up model convergence and improve classification performance.

3) Impact of Iterative Training : The proposed method uses the PFT at the $t-1$ iteration to divide unlabeled samples into reliable unlabeled samples and unreliable unlabeled samples. To demonstrate that IT can improve the discrimination of the proposed method, we have conducted some experiments on PUSL without IT. PUSL without IT means that PUSL uses the PFT obtained in the first stage to divide the unlabeled samples into reliable and unreliable unlabeled samples. Table V records the OAs of PUSL and PUSL without IT in the experiment using ten labeled samples per category for training. PUSL without IT weakens the discrimination of the network and is not beneficial to optimize the network parameters. With the deepening of model training, the discrimination of PUSL will continue to improve. Therefore, we use an IT strategy that uses the PFT obtained at the $t-1$ iteration to divide unlabeled samples into reliable and unreliable unlabeled samples.

F. Parameter Analysis

1) Impact of Position Encoding : Several experiments are conducted to verify the effect of PE on the performance of the backbone PFT and the proposed PUSL. PFT, PFT with PE, PUSL, and PUSL with PE are used to conduct classification experiments, respectively. PFT means only using labeled samples to optimize the network parameters of PFT in a supervised training manner. PFT with PE represents that both the labeled samples and PE information are used for supervised training. PUSL with PE means that the PE information is also used to optimize the network parameters in a semi-supervised training manner. Table VI records the experimental results of using ten labeled samples per category for training. Compared with PUSL, the OAs of PUSL with PE are decreased by 20.33%, 15.86%, and 6.11% on the IP, KSC, and Houston2013 datasets, respectively. When the number of training samples is small (for example, there are only ten training labeled samples per category for training), the spatial distribution diversity of training samples is also small. Inputting the PE information of pixels into PFT will make the proposed method pay much attention to the spatial distribution of labeled training samples, which can lead to overfitting the spatial distribution of labeled training samples and reducing the generalization ability.

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TABLE VII
OAs (%) of PUSL and Compared Methods With Five Labeled Samples per Category for Training

Database	SSAN	SpectralFormer	MDL4OW	DCFSL	CASSN	SSFTT	SS-CNN	SSGPN	3DAES	X-GPN	PUSL
IP	52.14	37.80	59.49	68.33	54.17	54.70	53.07	47.12	62.04	59.09	**72.16**
KSC	75.37	65.91	70.39	**88.52**	76.35	71.41	85.47	76.35	76.32	84.10	88.31
Houston2013	68.50	64.89	75.17	76.60	69.01	65.62	70.74	64.33	69.53	70.88	**78.01**

TABLE VIII
OAs (%) of PFT With Pixel Tokens and Patch Tokens

Database	Methods	10	20	30
IP	PFT with Patch Tokens	79.06	87.27	90.19
	PFT with Pixel Tokens	**79.65**	**87.52**	**90.27**
KSC	PFT with Patch Tokens	92.16	**96.79**	**98.03**
	PFT with Pixel Tokens	**92.84**	96.27	97.90
Houston2013	PFT with Patch Tokens	85.36	90.17	91.71
	PFT with Pixel Tokens	**86.35**	**91.68**	**92.58**

2) Impact of Patch Sizes: Several experiments are conducted to explore the performance of PUSL under different spatial sizes of HSI patches. Fig. 8 records the experimental results on three datasets. Since the number of adjacent pixels contained in a small HSI patch is limited, a small number of adjacent pixels cannot effectively provide sufficient spectral–spatial information for classification. As shown in Fig. 8, when the spatial size of HSI patches is set to 3, the OA of PUSL is low on three datasets. As the spatial size of HSI patches gradually increases, HSI patches can provide richer spectral–spatial information, which is beneficial to improve classification performance. However, the OA of PUSL does not always improve as the spatial size of HSI patches becomes larger. A larger HSI patch can also introduce more adjacent pixels of different categories. Adjacent pixels with different categories can interfere with the performance of PUSL.

3) Impact of Hyperparameters $\lambda_{u,r}$ and $\lambda_{u,u}$: Several experiments are conducted to explore the effect of different hyperparameters $\lambda_{u,r}$ and $\lambda_{u,u}$ on the performance of PUSL. Fig. 9 records the experimental results of using ten labeled samples per category for training. The value range of hyperparameters $\lambda_{u,r}$ and $\lambda_{u,u}$ is set to {0.001, 0.01, 0.1, 0.5, 1.0, 1.5, 2.0, 2.5}, respectively. As shown in Fig. 9, when the values of $\lambda_{u,r}$ and $\lambda_{u,u}$ are set to 0.1 and 0.5, the proposed PUSL achieves the best performance on the IP dataset. When the values of $\lambda_{u,r}$ and $\lambda_{u,u}$ are set to 0.5 and 0.1, the proposed PUSL achieves the best performance on the KSC and Houston2013 datasets.

4) Impact of Initial Ratio: PUSL regards the class confidence of unlabeled samples higher than the threshold T as reliable unlabeled samples. The threshold T affects the number of reliable unlabeled samples and unreliable unlabeled samples, which in turn affects the performance of PUSL. Since the threshold T in the proposed PUSL is determined by the initial ratio γ_0, we change the initial ratio to conduct some experiments to analyze the impact of different thresholds T on the performance of PUSL. Fig. 10 records the OA of PUSL at different initial ratios in the experiment using ten labeled samples per category for training. A small initial ratio means that the threshold T is large at the initial stage of the model training, which results in filtering out more reliable unlabeled samples. As the initial ratio increases, the OA of PUSL on three datasets also improves. However, the OA does not continuously improve as the initial ratio becomes larger. A too-large initial ratio can also cause the reliable unlabeled samples to introduce more wrong labels, which can weaken the performance of PUSL. PUSL sets the initial ratio to 80% for experiments on different datasets.

5) Performance of Different Methods With Few Labeled Samples: Obtaining sufficient labeled samples is time-consuming. Several experiments are conducted to further explore the classification performance of PUSL and compared methods in the experiment using five labeled samples per category for training. Table VII records the OAs of different methods. SSAN, SpectralFormer, CASSN, and SSFTT are prone to overfitting due to few labeled samples. For example, on the IP dataset, the OAs of SSAN, SpectralFormer, CASSN, and SSFTT are 52.14%, 37.80%, 54.17%, and 54.70% in the experiment using five labeled samples per category for training. Compared with SSAN, SpectralFormer, CASSN, and SSFTT, DCFSL overcomes the domain shift to improve classification performance. On the Houston2013 dataset, the OA of DCFSL on five labeled samples per category for training improved by 8.10%, 11.71%, 7.59%, and 10.98%, respectively. PUSL can make full use of reliable unlabeled samples and unreliable unlabeled samples and achieves the best performance in terms of OA, AA, and kappa on the IP and Houston2013 datasets.

6) Impact of Pixel Tokens and Patch Tokens: Although the patch tokens will facilitate the extraction of local features, overlapped sampling of patch tokens can also bring redundant information, which can affect the discrimination of features. We conduct some experiments to analyze the performance of PFT with pixel tokens and PFT with patch tokens. In the experiment of PFT with patch tokens, we crop hw overlapped patch tokens $X_i^p \in \mathbb{R}^{3\times3\times D}$ from the HSI patch $X_i \in \mathbb{R}^{h\times w\times D}$. The flattening operation and the fully connected operations are used to generate the embeddings of patch tokens. Then, hw embeddings of patch tokens and a learnable class embedding token are used as the input of PFT to extract the spectral–spatial features. As shown in Table VIII, PFT with pixel tokens achieves better results on both the IP and Houston2013 datasets. On the KSC dataset, the performance of PFT with pixel tokens and PFT with patch tokens is close. Therefore, the proposed method directly takes pixel tokens as input.

G. Time Cost

The time cost of PUSL and compared methods on the IP, KSC, and Houston2013 datasets is reported in Table IX. Since SSAN, SpectralFormer, MDL4OW, CASSN, and SSFTT only use a small number of labeled labels for training, the training time of these methods is relatively low. The training

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TABLE IX
TIME COST (SECONDS) OF DIFFERENT METHODS ON TEN LABELED SAMPLES PER CATEGORY

Datasets	Methods	SSAN	SpectralFormer	MDL4OW	DCFSL	CASSN	SSFTT	SS-CNN	SSGPN	3DAES	X-GPN	PUSL
IP	Training time	64.21	25.92	66.21	1069.61	20.26	35.21	636.23	64.89	1,028.38	129.28	943.44
	Inference	1.01	2.57	1.08	1.48	2.11	0.45	1.36	0.81	0.26	0.46	0.78
KSC	Training time	56.74	47.09	55.25	2308.51	14.5	46.53	366.89	29.48	698.51	60.78	607.71
	Inference	0.51	1.09	1.14	1.48	0.89	0.64	0.6	0.48	0.18	0.32	0.43
Houston2013	Training time	75.31	49.81	43.48	1404.11	16.72	61	852.92	55.19	1,177.89	101.01	1078.08
	Inference	1.75	1.80	2.01	4.41	2.15	0.87	1.49	0.74	0.17	0.75	0.97

time of 3DAES on the pretraining autoencoder stage is much longer than the time spent on fine-tuning the classifier stage. Since the SSGPN and X-GPN methods use a graph method to represent the topological information of HSI, the size of the adjacency matrix is reduced, which is beneficial to reduce computing resources. SS-CNN, 3DAES, and PUSL aim to make full use of a large number of unlabeled samples to improve classification performance, and the training time of these methods is higher than that of SSAN, SpectralFormer, MDL4OW, CASSN, and SSFTT. Since PUSL uses reliable samples and unreliable unlabeled samples to optimize network parameters at the same time, the proposed method requires a shorter training time compared with SS-CNN and 3DAES.

H. Discussion

To explore the effect of unreliable unlabeled samples, we conduct extensive experiments and analysis on public datasets, which demonstrate that unreliable unlabeled samples can be used to improve classification discrimination. The proposed PUSL exploits the features of unreliable unlabeled samples by contrastive learning to improve performance. In addition, different from previous methods, PUSL removes the PE module in PFT to reduce the interference of spatial distribution of labeled samples on extracting discriminative features. However, since the small number of labeled training samples can limit the performance of PUSL, the reliable unlabeled samples selected by PUSL may still contain wrong labels. Wrong labels can cause the network to learn inaccurate features and lead to error accumulation, limiting classification performance. Therefore, there is still much room for improvement in the quality of pseudolabels.

V. CONCLUSION

In this article, we propose a PUSL method for semi-supervised HSI classification. PUSL aims to make full use of numerous unreliable unlabeled samples for training. First, a PFT is built as the backbone to avoid overfitting the spatial distribution of labeled samples. Then, reliable unlabeled samples are used together with labeled samples to optimize PFT in a supervised learning manner. Next, unreliable unlabeled samples are used as negative samples for contrastive learning to further improve the discrimination of PFT. Extensive experiments on three HSI datasets demonstrate the effectiveness of PUSL. In future, we will investigate lightweight deep networks that directly explore unlabeled samples for learning discriminative features for HSI classification.

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Huaxiong Yao received the bachelor's and Ph.D. degrees in information and communication system from the Huazhong University of Science and Technology, Wuhan, China, in 2001 and 2007, respectively.

He is currently with the School of Computer Science, Central China Normal University, Wuhan. His research interests include deep learning and computer vision.

Renyi Chen received the B.Eng. degree in electronic information engineering from the Tianjin University of Technology and Education, Tianjin, China, in 2021. He is currently pursuing the M.E. degree in computer science and technology with the School of Computer Science, Central China Normal University, Wuhan, China.

His research interests include computer vision, spatial–temporal data analysis, and deep learning.

Wenjing Chen received the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2021.

She is currently a Lecturer with the School of Computer Science, Hubei University of Technology, Wuhan, China. Her research interests include deep learning and hyperspectral image processing.

Hao Sun (Member, IEEE) received the Ph.D. degree in signal and information processing from the University of Chinese Academy of Sciences, Beijing, China, in 2021.

He is currently a Lecturer with the School of Computer Science, Central China Normal University, Wuhan, China. His research interests include computer vision, deep learning, and hyperspectral image analysis.

Wei Xie is currently a Professor with the School of Computer Science, Central China Normal University, Wuhan, China. His research interests include image processing, computer vision, and deep learning.

Xiaoqiang Lu (Senior Member, IEEE) received the Ph.D. degree in signal and information processing from the Dalian University of Technology, Dalian, China, in 2010.

He is currently a Full Professor with the College of Physics and Information Engineering, Fuzhou University, Fuzhou, China. His research interests include intelligent optical sensing, pattern recognition, machine learning, and hyperspectral image analysis.

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